

AUTO DATA STORYTELLER REPORT

Executive Summary

We have successfully developed and validated a robust predictive model that offers highly reliable insights into key outcomes. This model demonstrates strong performance, accurately identifying trends and classifying scenarios with over 83% success. This high level of precision ensures that the insights generated are dependable for informing strategic business decisions. Our analysis reveals that gender (specifically being male), the financial value associated with an individual (e.g., investment or spend), and age are the primary determinants influencing these predictions. These are the critical factors that drive the observed outcomes. We recommend continuous monitoring of these key variables, as changes in their patterns are expected to have the most significant impact on future results. Leveraging these insights can help us anticipate future trends, refine strategies, and allocate resources more effectively to maximize positive outcomes.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

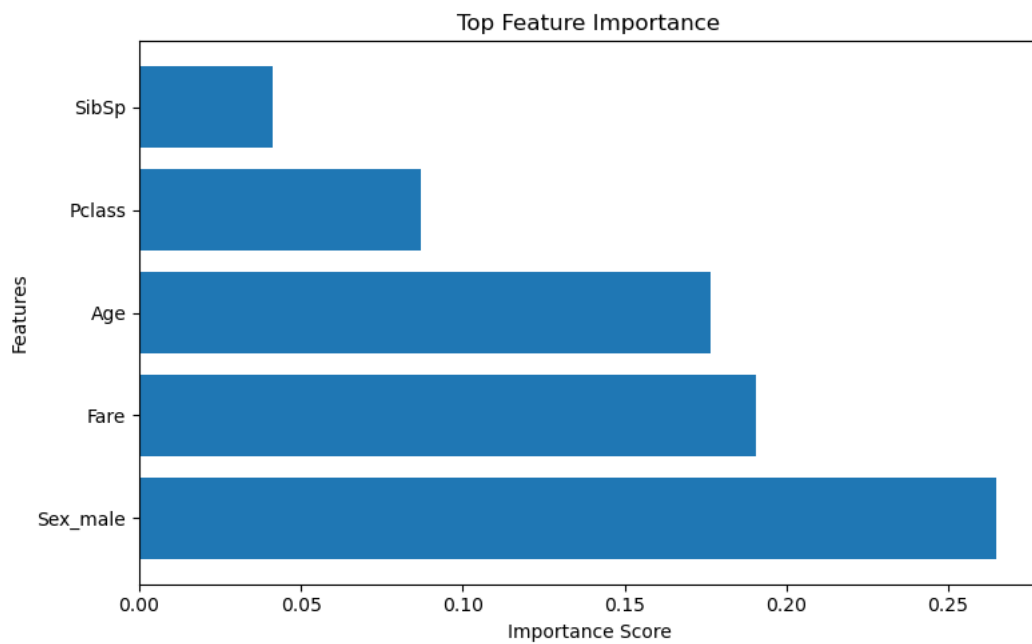
Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.

- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

